

PYTHON FOR PLATFORM ENGINEERS · ISSUE 02 OF 04

Health checks for APIs and Kubernetes in Python.

A check without a timeout is just another thing that hangs.

02

AR

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Swipe →

What a useful check covers

01 Status and latency

200 OK within a budget, not just a response.

02 Dependencies

The endpoints the service needs to work.

03 TLS expiry

Days left on the certificate.

04 Deployment state

Available replicas vs desired.

HTTP with timeouts and retries

```
import requests
from requests.adapters import HTTPAdapter
from urllib3.util.retry import Retry

retry = Retry(total=3, backoff_factor=0.5,
              status_forcelist=[502, 503, 504])
s = requests.Session()
s.mount("https://", HTTPAdapter(max_retries=retry))

r = s.get("https://api.example.com/healthz",
          timeout=(3, 10))
ok = r.ok and r.elapsed.total_seconds() < 1.0
```

Deployment health

```
from kubernetes import client, config

config.load_incluster_config()
apps = client.AppsV1Api()

for d in apps.list_namespaced_deployment("payments").items:
    want = d.spec.replicas or 0
    have = d.status.available_replicas or 0
    if have < want:
        print(f"DEGRADED {d.metadata.name} {have}/{want}")
```

Run it with a read-only service account scoped to the namespace.

Exit codes for automation

CODE	MEANING
0	Healthy
1	Degraded — alert
2	Check itself failed

Checker habits

DO

- ✓ Connect and read timeouts
- ✓ Separate degraded from broken checker
- ✓ Read-only RBAC

AVOID

- ✗ Checks that only test a port
- ✗ No timeout
- ✗ Cluster-admin for a health check

KEY TAKEAWAY

Time-box every call and make the exit code mean something.

- Check latency, dependencies, TLS, replicas
- Retries with backoff, bounded
- Read-only identity

YOUR TURN

What does your post-deploy smoke test actually check?

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Next in the series: #03 Python validation scripts in CI

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